

Atharva Niranjan Joshi

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Professional Summary

Highly accomplished Technical Lead and Senior Data Scientist specializing in cutting-edge Natural Language Processing (NLP), Generative AI, and Large Language Models (LLMs). Proven track record of architecting and deploying production-grade AI systems, multi-agent frameworks, and advanced Retrieval-Augmented Generation (RAG) solutions within highly regulated domains, specifically Trader Surveillance, Financial Misconduct Detection, and Call Center Analytics. Expert in unstructured data processing (voice, eComms, OCR), LLMops/MLOps pipelines, and scalable cloud/on-premise deployment architectures utilizing the Azure AI ecosystem and self-hosted open-source models. Strong leader adept at guiding agile engineering teams from ideation to enterprise delivery.

Technical Skills

- **Generative AI & Advanced NLP:** Prompt Engineering & Optimization, Fine-Tuning (LoRA, QLoRA), Model Context Protocol (MCP), Multi-Agent Orchestration (LangGraph, LangChain), LlamaIndex, vLLM, Open-Source LLMs (Llama, Mistral) via Ollama, OpenAI APIs, Tokenization, Embeddings, BERT, NLTK, SpaCy.
- **Search & Vector Architectures:** Advanced RAG, Azure AI Search, ChromaDB, Vector Databases, Hybrid Search, Reciprocal Rank Fusion (RRF), Cross-Encoder Re-ranking, Semantic Search, Knowledge-Grounded Reasoning.
- **Voice Analytics & Document Intelligence:** Azure Speech Services (Batch & Real-Time APIs), Speaker Diarization, Audio Preprocessing, Azure AI Document Intelligence (OCR), PyMuPDF, Unstructured Data Ingestion (Emails, Chats, Voice).
- **LLMops, MLOps & Governance:** Ragas, NeMo Guardrails, Llama Guard, MLflow, CI/CD Pipelines, Prompt Versioning, Automated Testing, Hallucination Detection, Responsible AI.
- **Cloud, Infrastructure & Data:** Azure Ecosystem (Azure OpenAI Service, Azure ML Studio, Azure Functions, App Services, Storage), Microsoft Fabric (Data Lakehouse, Warehouse), Docker, Kubernetes, AWS (S3), REST APIs, SQL, Python (PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy).

Work Experience

Technical Lead & Data Scientist (Assistant Manager) | KPMG, UK

Oct 2022 – Present

- **Enterprise GenAI & Agentic AI Architectures:** Serve as the Technical Lead for multiple enterprise Generative AI initiatives. Designed and deployed a multi-agent framework utilizing the Model Context Protocol (MCP) and LangGraph to build a suite of specialized AI agents for intent/entity classification, achieving 100% QA coverage in detecting complex financial and non-financial risks across eComms and voice channels.

- **Advanced RAG for Financial Crime Investigation:** Engineered a production-grade Advanced RAG solution leveraging Azure AI Search and ChromaDB to ingest massive volumes of conversational data. Implemented hybrid search, Reciprocal Rank Fusion (RRF), and cross-encoder re-ranking, allowing compliance teams to pinpoint exact risky communications and saving dozens of hours previously spent manually reviewing raw audio and eComms datasets.
- **Voice Analytics & SLA Optimization:** Overhauled and scaled end-to-end voice processing solutions using Azure Speech-to-Text (Batch & Real-Time). Engineered custom preprocessing, diarization, and downstream LLM evaluation components, slashing system SLAs from 3 days down to 1 day for large-scale call center analytics.
- **Open-Source Optimization & On-Prem Deployment:** Led fine-tuning strategies (LoRA and QLoRA) to adapt open-source LLMs to highly specialized regulatory domains. Orchestrated secure offline/on-premise model deployments using Ollama and vLLM with AWQ/GGUF quantization techniques to ensure data residency compliance.
- **Document Intelligence & Enhanced OCR:** Designed advanced unstructured data pipelines utilizing Azure AI Document Intelligence and custom OCR preprocessing. Significantly reduced data extraction leakage on low-quality PDFs and financial communications, increasing downstream risk-capture rates by an estimated 25% due to high-fidelity text extraction.
- **LLM Evaluation, Safety & Guardrails:** Designed comprehensive LLM governance and evaluation frameworks utilizing Ragas for factual accuracy and retrieval quality. Implemented enterprise-grade validation layers with NeMo Guardrails to mitigate prompt injection, ensure structured outputs, and filter non-compliant content.
- **Legacy System Modernization & Hybrid ML:** Supercharged legacy NLP systems by blending traditional ML (TF-IDF, BERT, Logistic Regression, PyTorch) with modern GenAI, resulting in a 70% reduction in false-positive outcomes and dramatically minimizing manual review overhead.
- **Infrastructure & Automated LLM Ops:** Automated cloud infrastructure deployments utilizing Docker, Kubernetes, and Azure Web Apps (Chainlit interfaces). Built end-to-end CI/CD and data engineering pipelines via Microsoft Fabric (Data Lakehouse), saving over 10 hours of manual work every week.
- **Team Leadership:** Coordinate Agile delivery cycles, conduct rigorous technical code/design reviews, manage stakeholder expectations, and mentor junior data scientists and machine learning engineers.
- *Key Achievement: Recognized with the "Employee of the Month" award for delivery excellence and pioneering technical contributions.*

Senior Data Analyst | Centre for Health Research and Education, UK

Jun 2021 – Aug 2022

- **Computer Vision & Deep Learning:** Researched and applied advanced deep learning image object detection and classification architectures (YOLOv4, Faster R-CNN) utilizing PyTorch to improve oral cancer diagnostic models.
- **Data Pipeline Management:** Coordinated with international medical experts to establish a secure data curation workflow. Managed storage architecture using AWS S3 and leveraged Labelling tools to build highly annotated image datasets for clinical model training.

Systems Engineer | Infosys Limited, India

Sep 2018 – Sep 2020

- **Cross-Platform Engineering:** Developed enterprise-grade mobile applications within the Xamarin.Forms framework using C# and XAML, leveraging Azure Active Directory for secure IAM authentication and Azure APIs for enterprise data retrieval.
- **Agile Development & Client Deliveries:** Spearheaded full-lifecycle sprint feature developments and issue resolution workflows via Azure DevOps. Successfully delivered high-stakes proof-of-concepts (PoCs) and pilot projects directly to institutional clients.
- *Key Achievement: Awarded a Certificate of Appreciation for outstanding performance and project execution.*

Education

- MSc in Big Data Science with Industrial Experience (Distinction – 86.31%) | Queen Mary University of London
- Post Graduate Diploma in Data Science (Grade: A) | IMS ProSchool, India
- Bachelor of Engineering (B.E.) in Information Technology (First Class) | Savitribai Phule Pune University

Certifications & Professional Compliance

- Generative AI with Large Language Models (Coursera)
- Natural Language Processing with Classification and Vector Spaces (Coursera)
- International Compliance Association (ICA) Certifications: Financial Crime Prevention, Conduct Risk, KYC and CDD, Managing Sanctions Risk
- Infosys Certified Python Programmer & Associate